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Investigation of Bound and Free Water in Plant-Based Food Material Using NMR T₂ Relaxometry

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Abstract

Plant-based food materials are porous and hygroscopic in nature; therefore, it contains three water environments, namely, intercellular, intracellular water and cell wall water. The intercellular water is known as capillary water or free water which is less constrained than intracellular water, considered as loosely bound water (LBW), and cell wall water, which is recognised as strongly bound (SBW). During food processing such as drying, frying, heating and cooking, optimisation of heat and mass transfer is crucial. The existing heat and mass transfer models for food processing are developed based on the concept that all of the water inside the food material is bulk water, which can act as free water that can be easily transported. This simplistic assumption has been made due to a lack of sufficient data to enable consideration of the proportion of free and bound water in plant-based food materials. Therefore, the aim of the present study is to investigate the proportion of different types of water such as free, LBW and SBW in 11 different plant-based food materials. The water proportion was investigated using ^1H -NMR T_2 relaxometry. The experimental results uncover that plant-based food materials contain about 80 to 92 % LBW, 6 to 16 % free water and only about 1 to 6 % SBW. This investigation also confirms that among the five different fruits, kiwi contains the lowest percentage of LBW while Apple contains the highest percentage of LBW. Among the vegetables, eggplant comprises the largest amount of LBW while cucumber contains least amount of SBW. An attempt was made to establish a relationship between physical properties of fruits and vegetables and the proportion of the different types of water. Interestingly, it was found that SBW strongly depends on the proportion of solid in the sample tissue whereas FW depends on the porosity of the material.

Keywords: Strongly bound water, loosely bound water, free water, plant-based food material, food processing

1. Introduction

Plant-based foods contain about 80-90 % water and are complex in nature due to their porous and hygroscopic nature. This vast amount of water exists in different spaces inside the hygroscopic plant based food tissue. Based on the spatial distribution of water, three types of water environment are recognised, namely capillary (intercellular) water, intracellular water and cell wall water. The intracellular and cell wall environmental water are recognised as physically bound water. On the other hand, a small proportion of water bound with the chemicals (nutrition) and is known as chemically bound water that should not be removed during food processing due to its great impact on the taste and flavour of the dried food (Kuprianoff, 1958). Therefore, only physically bound water is the main concern for transport during food processing (Khan et al., 2016 a) that is discussed below.

Water residing in the intercellular space is known as capillary water or free water (FW) (Joardder et al., 2015c). The water inside the cell is referred to as intracellular water and the cell wall water is that water occupying the fine space inside the cell wall. The latter two types of water (intracellular water and cell wall water) are known as physically bound water or simply bound water (Karel, 2003). Based on the mobility of water, bound water present inside the cells (intracellular water) is referred to as loosely bound water (LBW) while cell wall water is termed as strongly bound water (SBW) (Caurie, 2011).

The plant-based food materials that are subjected to drying processes can be treated as hygroscopic, porous and amorphous media which undergo multiphase transport of heat and mass (Srikiatden and Roberts, 2007). Most of the existing food drying models consider all water inside the food material as transportable i.e. bulk water. (Chemkhi et al., 2009; Yamsaengsung and Moreira, 2002; Datta, 2007; Halder and Datta, 2012; Feyissa et al., 2013;

van der Sman, 2013; Gulati and Datta, 2015; Kumar et al., 2016). During this type of modelling, all the water is considered as intercellular (free water); therefore, it does not consider water inside the solid matrix. Due to the hygroscopic nature of plant-based food materials, bound water transport has a great effect on material shrinkage during food drying. Joardder et al. (2015a) suggested that migration of free water has a minor effect on material structure, whereas migration of LBW contributes to cellular shrinkage, pore formation and the collapse of the cell. Furthermore, overall food tissue is deformed due to the movement of SBW. This is consistent with the report by Prothon et al. (2003) that showed migration of three types of water is responsible for overall tissue shrinkage due to cellular shrinkage as well as structure collapse. Therefore, without considering bound water transport, a food drying model may not provide realistic understanding of the heat and mass transfer mechanism as well as material deformation (shrinkage) during drying.

An understanding of the proportions of free and bound water in a fresh sample is crucial information for developing microscale food processing model. Few researchers have attempted to investigate the characteristics of different water environments in plant-based food material. Therefore, their conceptual approach does not provide the data needed for modelling of multiple compartments. Aguilera and Stanley (1999) quantified the percentage of free water present in carrot, potato and apple tissues; however, they did not investigate the proportion of LBW and SBW in those tissues. The literature indicates that at least 70% of the water is intracellular in meat (Honikel and Hamm, 1994) and about 74% in vegetables (Asquith et al., 2007; Nobel, 2009); however, there are no reports of the proportions of water in the other two types of water environments. Therefore, measurement of the accurate proportions of FW, LBW and SBW in plant-based food material will contribute to the development of accurate heat and mass transfer model.

1 There are several methods that can potentially be used for measuring the different types
2 of water, including differential scanning calorimetry (DSC), differential thermal analysis
3 (DTA), dilatometry, bioelectric impedance analysis (BIA), and nuclear magnetic resonance
4 (NMR).

5 DSC is a very informative method for the physical characterisation of a compound in a tissue
6 or material. It can monitor the heat effects associated with phase transitions and chemical
7 reactions as a function of temperature. Using a DSC curve (temperature vs. heat flow), the
8 different composition of plant tissue can be predicted. The DSC and DTA methods are quite
9 similar. Many researchers used this technique for investigating the bound water for different
10 materials, for instance, sludge (Lee and Hsu, 1995), carboxymethylcellulose (Nakamur et al.,
11 2004), polymer (Hatakeyama et al., 1988), cotton, kapok, linen, jute, wood (Nakamura et al.,
12 1981), and beef (Aktaş et al., 1997; Schwartzberg, 1976). The DSC and DTA techniques can
13 provide only the qualitative information. Goni et al. (2011) investigated the effect of water
14 binding capacity on the quality of food material using DSC. They determined the amount of
15 unfreezable water and its effect on quality retention in high-CO₂ treated table grapes.
16 However, due to the inability to obtain quantitative information on water distribution in the
17 tissue, they could not investigate the proportion of FW, LBW, and SBW.

18 The dilatometry technique is the approach to investigate bound water based on the volume of
19 freezable and unfreezable water expansion and contraction, respectively. This method is
20 basically suitable for investigation of bound water in different non-hygroscopic material such
21 as soil (Buehrer and Rosenblum, 1939), wastewater (Smith and Vesilind, 1995), and sludge
22 (Lee, 1996). However, this method is not suitable for investigation of bound water in plant
23 based food material due to its hygroscopic in nature.

24 Bioelectrical Impedance Analysis (BIA) is a very simple and old technique for measuring
25 body composition. It measures the resistance of tissues to the flow of electrical current. The

proportion of different composition (water or fat) in tissue can be calculated as the current flows more easily in the parts of the tissue that are composed mostly of water. Researchers are trying to use this technique for measuring cellular water in plant-based food material. Halder et al. (2011) investigated the percentage of intracellular water in plant-based food materials using BIA. However, they only investigated the proportion of intracellular water; ignoring the percentage of FW and SBW. The main limitation of the technique is that it cannot locate the position and distribution of different types of water within the tissue.

Proton nuclear magnetic resonance (^1H -NMR) T_2 relaxation method has proven to be valuable in the study of plants and plant-based food materials submitted to stress, reflecting anatomical details of the entire tissue based on the water distribution in particular (Van Der Weerd et al., 2002; Gambhir et al., 2005). ^1H -NMR relaxation signals, which are an average over the whole sample, provide information on the water environments within the plant tissue since the proton signal from the plant tissue is dominated by water protons (van der Weerd et al. 2001) and the proton NMR signal intensity is directly proportional to the proton density of the tissue (Westbrook, 1993). The water proton relaxation behaviour strongly depends on the water mobility in the microscopic environment of the tissue, local magnetic field fluctuations (related to the molecular environment) and the strength of the applied magnetic field. The spin-spin (T_2) relaxation is the transverse component of the magnetization vector, which exponentially decays towards its equilibrium value after excitation by radio frequency energy. It can be expressed as

$$M(t) = \sum_{i=1}^n A_i \times \exp(-t / T_2^i) \quad (1)$$

where $M(t)$ is the function of relaxation time, A is the relative contribution of sets of protons, T_2 is the relaxation time of water proton, and i is the number of contributing component.

Deconvolution of the T_2 signal can show contributions from water in different environments, which represents the differences in molecular mobilities of water in different macromolecular systems (Deslauriers and Smith, 1980). Using this technique, many studies have been conducted to investigate the different water components in animal tissues such as free and bound water in animal lung (Cutillo et al., 1992; Sedin et al., 2000), brain (Furuse et al., 1984; Inao et al., 1985; Berenyi et al., 1998; Vajda et al., 1999; Sulyok et al., 2001), liver (Moser et al., 1996; Moser et al., 1992), and red blood cells (Besson et al., 1989).

In the case of plant-based food material, T_2 relaxation measurements have been applied to the investigation of sugar content in fruit tissue (Goni et al., 2013; Arus, 2013), quality of fruits and vegetables (Chen, 1989; Velde et al., 2016), and the maturity of fruits and vegetables (Chen et al., 1993; Ruan et al., 1999). Literature describing measurement of free and bound water using this method in plant-based food materials is relatively sparse. Hills and Remigereau (1997) investigated T_2 relaxation times for assessing the migration of water during drying in apple tissue. However, they did not quantify the various types of water in that tissue. Gonzalez et al. (2010) investigated the change in T_2 with processing time the effect of high pressure on cell membrane integrity during thermal processing of onion. However, they did not investigate the proportion of FW, LBW and SBW in the onion tissue.

Furthermore, in considering the special attention on post-harvesting quality assurance, much research has been conducted using NMR T_2 relaxation to investigate the water core (Clark et al., 1998; Cho et al., 2008; Herreros et al., 2013), internal browning (Clark and Burmeister, 1999; Gonzalez et al., 2001; Cho et al., 2008) and microstructural heterogeneity (Defraeye et al., 2013; Winisdorffer et al., 2015) in apple tissue. However, these studies did not investigate the proportion of FW, LBW and SBW considerations in the transportation of water during food processing like drying, frying, and baking.

1 Cho et al. (2008) investigated the role of different NMR-T₂ relaxation behaviour on the internal
2 browning and water core development of apple tissue under different conditions using the
3 multi-component analysis method. The main concern of Cho et al. was to investigate the post-
4 harvesting quality assurance of Apple tissue. Therefore, the classification of water
5 compartments was based on single time point of interest in different compartments such as the
6 vacuole, the cytoplasm and the cell wall together with the extracellular space.

7 Given that the majority of the protons in plant tissue are found in water, the predominant NMR
8 signal can be attributed to tissue water. T₂ relaxation time constants are sensitive to the local
9 milieu, with more restrictive environments in which molecular motion is slowed, yielding
10 shorter T₂ relaxation times. When the tissue being examined has multiple environments, the
11 measured NMR signal will be a weighted average of the signal from the individual tissue
12 compartments. The approach used in this study is different from others in that we have used an
13 imaging method to obtain data from selected regions of tissue, whereas others have used high
14 resolution NMR methods that can only provide information that is an average from the entire
15 sample. In a homogenous tissue, this would provide equivalent data; however, when there are
16 structural variations within the sample, the T₂ map provided by imaging can target specific
17 tissue regions within the sample.

18 Therefore, the primary aim of this study was to experimentally investigate the proportion of
19 three types of water namely free water, LBW (intracellular water), and SBW (cell wall water)
20 for various plant-based food materials using ¹H NMR T₂ relaxation measurement and to
21 develop a relationship between different environmental water with the physical properties of
22 the food material.

23 Knowledge of the proportions of three types of water (free, LBW, and SBW) and the
24 relationship of this types of water with the physical properties will contribute to the
25 development of a more accurate heat and mass transfer model for use in food processing.

2. Materials and Methods

2.1. Sample Preparation

Experiments were performed for five different fruits namely, apple, pear, kiwi, nectarine, apricot and six different vegetables potato, eggplant, pumpkin, tomato, cucumber, and carrot. The samples were collected from the local market in Brisbane, Australia. Samples were stored in a refrigerator at 4 °C prior to NMR measurement. At the start of each experiment, the materials were washed and cut into cylindrical shapes of 30 mm height and 10 mm radius and then weighed on digital electronic balance (model BB3000; Mettler-Toledo AG, Grefensee, Switzerland) with an accuracy of 0.01 g.

2.2 Porosity measurement

Porosity (φ) of food material is determined by the ratio of particle volume and bulk volume as expressed by the following equation:

$$\varphi = 1 - \frac{V_p}{V_b} \dots\dots\dots (2)$$

where V_p and V_b are the particle volume and bulk volume of the food material.

In this study, a solid displacement method was used to measure the bulk volume (V_b) of the sample. Particle volume (V_p) was determined with the gas (helium) pycnometer (Quantachrome pentatype 5200e, USA). Helium is used due to its small molecular size, which makes it possible to access the smallest pores. Details of the porosity measurement procedure has been adopted from Authors' previous publications (Joardder et al., 2015b).

2.3 Calculation of solid material content

The percentage of solid content in each sample was calculated from the following equation

$$\text{Solid content} = 1 - \frac{W_i - W_{dry}}{W_i} \dots\dots\dots (4)$$

where W_i is the initial weight of the sample and W_{dry} is the weight of bone dried sample. To calculate the percentage of solid content in the sample, the initial weight of the sample and the weight of the sample at bone dried condition were taken. The samples were dried about 24 hours in a commercial hot air cabinet dryer to get a bone dried sample. Drying experiments were performed at the temperature of 60 °C and constant air velocity of 0.7 m/s. During the experiments, the relative humidity of the inlet air was 60%. It is reported that to obtain a material weight at bone dried condition, the sample should be dried about 24 hrs at the above drying conditions (Kumar et al., 2015).

2.4. NMR measurement

Spatial T_2 measurements were made with a Bruker DRX wide-bore 7 tesla magnet and spectrometer operating at 300 MHz for hydrogen, equipped with a micro-imaging (mico 120) gradient set (Bruker Bio-spin, Karlsruhe, Germany). Each prepared sample was placed in a 25 mm diameter NMR tube, sealed with a standard NMR tube cap and allowed to stand for 5 min to reach thermal equilibrium (22°C). The auto tune feature of the Paravision 4 software (version 4.2) was used to automatically adjust the shimming based on the water signal and the centre frequency was adjusted prior to data acquisition. Quantitative multi-echo transversal T_2 imaging was carried out using a Multi-Slice-Multi-Echo (MSME) sequence with the following acquisition parameters: repetition time 5000 ms, 128 equally spaced echoes between 10 and 1280 ms, 64 averages. The slice thickness, field of view, and the matrix size were 3 mm, 30mm and 64, respectively, giving an in-plane spatial resolution of 468 μm .

2.5. Mathematical analysis

2.5.1. Theory: The water mobility in different plant-based food tissue was investigated by assessing the water fractions quantitatively using multicomponent analysis of the T_2 relaxation decay curves. Generally, an exponential decay curve is followed by the free induction decay of the proton (predominantly from water) relaxation process. A multi-exponential equation can describe this function. Each tissue compartment corresponding to a different environment will have a distinct T_2 relaxation time constant, we assumed that these compartments were not inter-reliant at the time of the measurements. It was also assumed that the volume fractions were constant and that the influence of exchange rates between compartments and chemical shift effects were negligible. Like others who have used NMR to examine food, we have attributed a specific small number of discrete exponential decay components as representing tissue compartments with different water mobility. It may not provide an exact model for complex heterogeneous and hygroscopic materials, such as food, that are undergoing diffusion and exchange processes, although this assumption is necessary to simplify the analysis (Chinachoti and Vodovotz, 2000). In a plant-based food material (a heterogeneous system), spins could exist in a variety of different environments, giving rise to range of relaxation times. Therefore, the observed relaxation decay will result from the contributions from water signals in each compartment (Chinachoti and Vodovotz, 2000). The multi-exponential nature of the T_2 decay curve results from the individual decay curves for the different water compartments in the tissue with minimal exchange on the NMR time-scale. In the case of fast exchange of protons between tissue water and macromolecules, the water compartments would be symbiotic, with the water proton relaxation following a pattern of mono-exponential decay (Shioya et al., 1990; Cole et al., 1993).

2.5.1 Data analysis:

For each sample, two ROIs (regions-of-interest) were defined manually on a given slice of the mid transverse section from the MSME images, as shown in Figure1. The mean value of these ROIs was computed for each sample. The accuracy of the multi-exponential parameters considerably depends on the noise level; therefore a third ROI was assessed outside the sample for determining the signal-to-noise ratio (SNR, average signal intensity over the standard deviation of the noise). Noisy T_2 -signals were eliminated from the original signal, as described in authors' previous publication (Nagy et al, 2014), and data with appropriate SNR ($SNR > 5$) were used for parameter fitting. In order to achieve $SNR > 5$, a constant cut-off of $TE < 600$ ms was used for all datasets (Wheeler-Kingshott et al., 2004). The mean signal of the two ROIs at each echo time were measured and plotted as a function of time on a semi-log scale where curvature was determined (Boulby and Gunn, 2004). The number of inflection points was used to determine the number of T_2 components. The different T_2 relaxation times comprising the signal from each sample was determined by mono-, bi- and tri-exponentially fitting the mean signal intensity using the following equation.

$$I_{(t)} = \sum_i I_{0,i} \exp(-t/T_{2,i}) \dots \dots \dots (5)$$

where t represents the time, $I_{0,i}$ is proportional to the local spin density of the i^{th} signal component and $T_{2,i}$ represents the transverse relaxation time constant of the i^{th} signal component. For T_2 measurements, odd echoes were excluded to minimize error due to the influence of stimulated echoes (Maier et al. 2003). T_2 relaxation time data processing was carried out with a non-negative least squares algorithm using self-written program code in the curve fitting toolbox of Matlab® software (The MathWorks, Inc., Natick, MA). The experiments were replicated three times for each sample, and the average of these measurements was used for further analysis.

[Figure 1 can be inserted here]

2.6. Statistical analysis: Data are expressed as mean $\pm$ SD of the mean. For statistical analysis, least-squares linear regression analysis was used.

3. Results and discussion:

In this study, the bound water of different plant-based food material was investigated using ^1H -NMR relaxometry. Two types of plant-based food material were considered for this investigation namely, fruits and vegetables. Figures 2(a) and 2(b) show the T_2 relaxation decay curve for five different fruits and six different vegetables, respectively.

To compare the accuracy of the model, curves were fitted with bi-exponential and tri-exponential decay equations. The goodness of fit (R^2) was calculated for both two and three component models. It was found that the goodness of fit for tri-exponential fitting model was better than the bi-exponential fitting model. Therefore, tri-exponential fitting was used in this study. After fitting different T_2 relaxation intensity data, three different proton components of water (slow, medium and fast) were described (Table 1) with their corresponding standard deviations and goodness of fit.

[Figure 2(a) can be inserted here]

[Figure 2(b) can be inserted here]

Three T_2 signal components namely, slow, medium and fast were categorised as LBW, FW, and SBW respectively. This categorisation is mainly based on the pore size, water content, membrane permeability and the proton density within the sample (Boulby and

Gunn, 2004; Saeedi, 2012). Moreover, water mobility at the molecular level influences T_2 relaxation. The water mobility is defined as the ability of water molecules to rotate freely, as well as moving spatially with the result that T_2 is inversely proportional to rotational motion (which will be a reflection of inherent mobility) (Nelson, 2003).

It is reported that the water in a restrictive environment shows fast relaxation times (Boulby and Gunn, 2004; Wheeler-Kingshott et al., 2004) and T_2 is also short when water is strongly interacting with macromolecules (Rondeau et al., 2014). In plant-based food material, the cell wall is mostly composed of solid matter with very low water content. This structure acts to restrict translational motion and increase the correlation time of water (Boulby and Gunn, 2004). Therefore, the fast T_2 component is most likely to relate to cell wall water (SBW). Furthermore, it has been stated that the majority of water is present in intracellular spaces that act as a water reservoir with a predominantly water-water interface (Asquith et al., 2007; Nobel, 2009; Joardder et al., 2015c; Khan et al. 2016a; Halder et al., 2011; Khan et al. 2016b) where the water proton relax slowly (Berna, 2010), resulting in a longer T_2 relaxation. Among the three T_2 components fitted, the longest component had the greater water fraction. Based on the water content of the sample, it can be related to the intracellular water (LBW). The remaining component (medium) then relates to the free water (FW).

Figure 3(a) shows a comparison of the percentages of the different types of water in different fruits. Depending on the type of fruit and its cell structure, about 80-90% LBW, 2-5% SBW and 10-20 % FW was observed. Due to the diversity of food composition, cell structure and orientation and the solute content, the proportion of SBW, LBW and FW differs for various plant-based food materials.

It can also be seen from the illustration (Fig. 3a) that among the five different fruits, apple contains the highest proportion of LBW (about 90%) while kiwi fruit demonstrated the least

amount of LBW. These observations may be due to the solid proportion (cell wall) of kiwi fruit being thicker than apple tissue, resulting the number of cells per unit volume and the space inside each cell is much less than in apple. This result is in good agreement with the study of Halder et al. (2011), which found that about 90% of the water in apple tissue exists in the intracellular space.

On the other hand, the SBW determined for kiwi fruit is higher than that of other fruits. This may be solely due to its high proportion of solid material content, as shown in Figure 4(a). The SBW of different fruits is proportional to the amount of solid. Nectarine is an exception, compared to other fruits, which may be due to its diverse properties as illustrated in figure 4(a).

[Figure 3(a) can be inserted here]

[Figure 3(b) can be inserted here]

Likewise, Figure 3(b) shows the comparison of the proportion of each water environment for the various vegetables studied. Eggplant contains greatest amount (about 92%) of LBW, followed by cucumber and tomato. This result is consistent with the study of Halder et al. (2011). Depending on the amount of solid mass, cell wall water binding capacity, and cell structure, vegetables contain about 1.0- 6.0 % SBW. From Figure 4(b) it can be seen that potato contains the highest amount of solid and showed a maximum amount of SBW, totalling about 5.8 %.

With the exception of pumpkin, all of the vegetables followed nearly a linear relationship between the percentage of solid and the percentage of SBW (Fig. 4b). Pumpkin tissue comprises both cellular and fibrous structural components which may contribute to this discrepancy.

1 [Figure 4(a) can be inserted here]

2 [Figure 4(b) can be inserted here]

3
4 Figures 5(a) and 5(b) show the relationship between initial porosity and FW of various
5 fruits and vegetables, respectively. It can be seen that the percentage of free water shows an
6 inverse relationship with the porosity. In hygroscopic porous material, air and water mutually
7 stay within intercellular spaces (Woolley, 1983) such that highly porous material will contains
8 a higher amount of air inside intercellular space. From Figure 5(a), it can be observed that kiwi
9 fruit contains the highest amount of free water, consistent with it having the lowest porosity.
10 On the other hand, apple with the greatest porosity contains the least amount of free water. In
11 the case of the vegetables studied, except pumpkin (as it is an exception for the reason described
12 as earlier), potato contains the maximum amount of free water as compared to other vegetables;
13 while eggplant contains the least (Figure 5b). It may be due to the fact that porosity of potato
14 is almost zero while porosity of eggplant is significant.

15
16 [Figure 5(a) can be inserted here]

17 [Figure 5(b) can be inserted here]

18
19 Figure 6 shows an average generic proportion of different types of water in plant-based food
20 materials. It is observed that about 80-92% of LBW, 6-16% FW and only 1-6% SBW is present
21 in plant-based food material. **This result is consistent with the findings of Halder et al.**
22 **(2011), and Karmer (1955), where they used BIA and calorimetry for measuring the LBW**
23 **and SBW respectively. Using NMR, Berényi et al. (1996) investigated these three types of**
24 **water in animal tissue and their result is consistent with the result of current study.**

[Figure 6 can be inserted here]

5. Conclusion:

¹H NMR measurements have been performed to investigate the proportion of free water, loosely bound water and strongly bound water in five different fruits and six different vegetables. The experimental results based on multiple component analysis of T₂ relaxation data show that, the proportion of FW, LBW and SBW present in plant-based food materials is about 80-92%, 6-16 % and 1-5% respectively. Among the different fruits, apple showed the highest fraction of LBW. Depending on the solid portion and porosity of various fruit tissues, SBW and FW both were found highest in kiwi fruit. On the other hand, eggplant contained the largest fraction of LBW (92% ± 4) relative to other vegetables, followed by cucumber and tomato. This study also confirmed that there is close relationship between porosity with the free water measured in different fruits and vegetables. The findings of this study enhance the understanding of plant-based food tissue and may contribute to a better understanding of potential changes occurring during food processing. The findings will also contribute to developing more accurate heat and mass transfer models and prediction of deformation during food processing.

Future direction:

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Figure captions:

Figure 1: Region of interest (ROI) for T₂ analysis

Figure 2(a): The T₂ relaxation curve for different fruits

Figure 2(b): The T₂ relaxation curve for different vegetables

Figure 3(a): The percentage of different types of water in various fruits

Figure 3(b): The percentage of different types of water in various vegetables

Figure 4(a): The relationship between SBW and percentage of solid for fruits

Figure 4(b): The relationship between SBW and percentage of solid for vegetables

Figure 5(a): Effect of porosity on free water for fruits (results are mean $\pm$ SD)

Figure 5(b): Effect of porosity on free water for vegetables (results are mean $\pm$ SD)

Figure 6: The percentage of different types of water in plant-based food material

Figure 1:

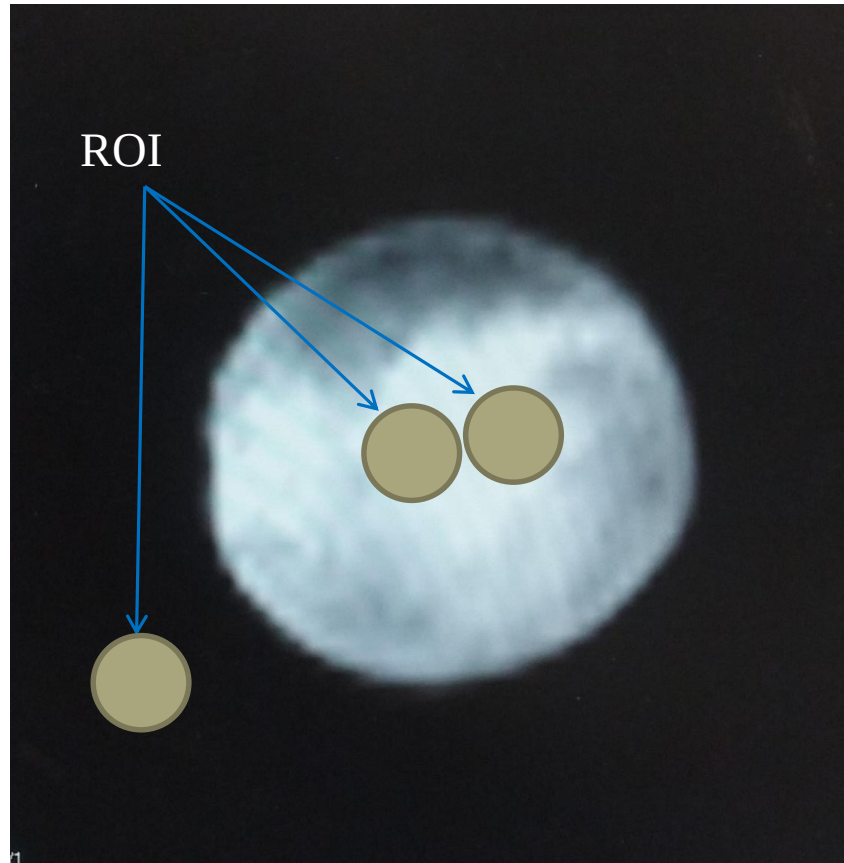


Figure 2(a):

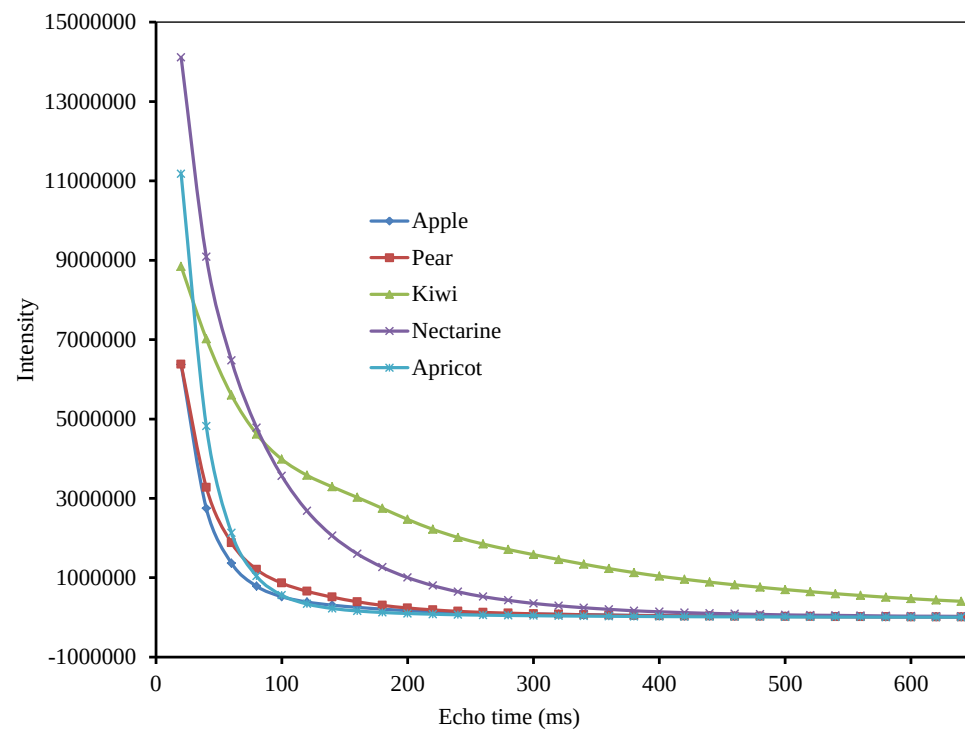


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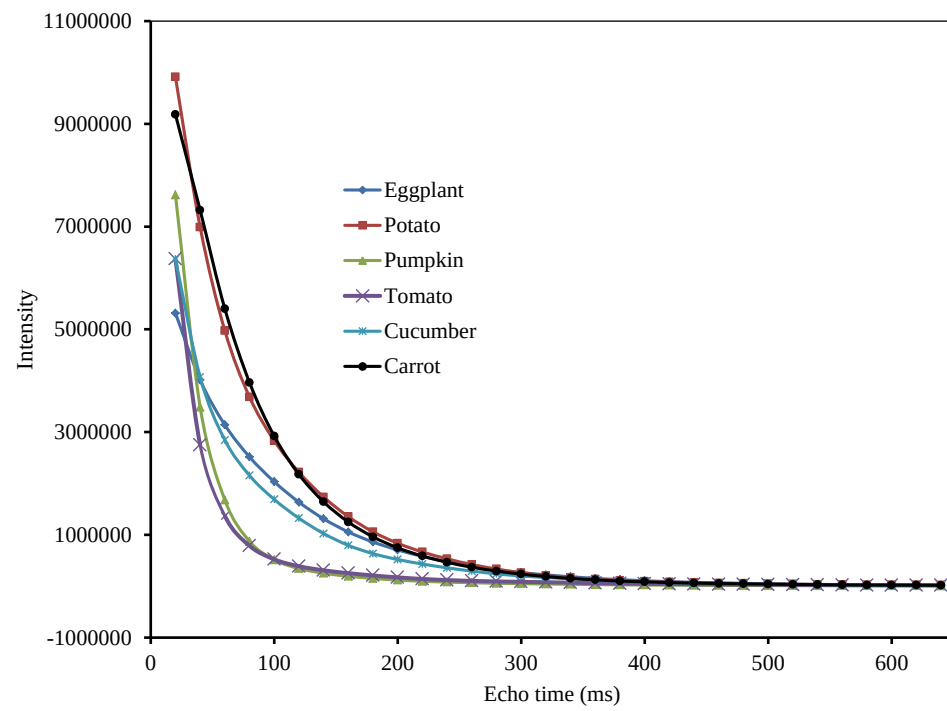


Figure 3(a):

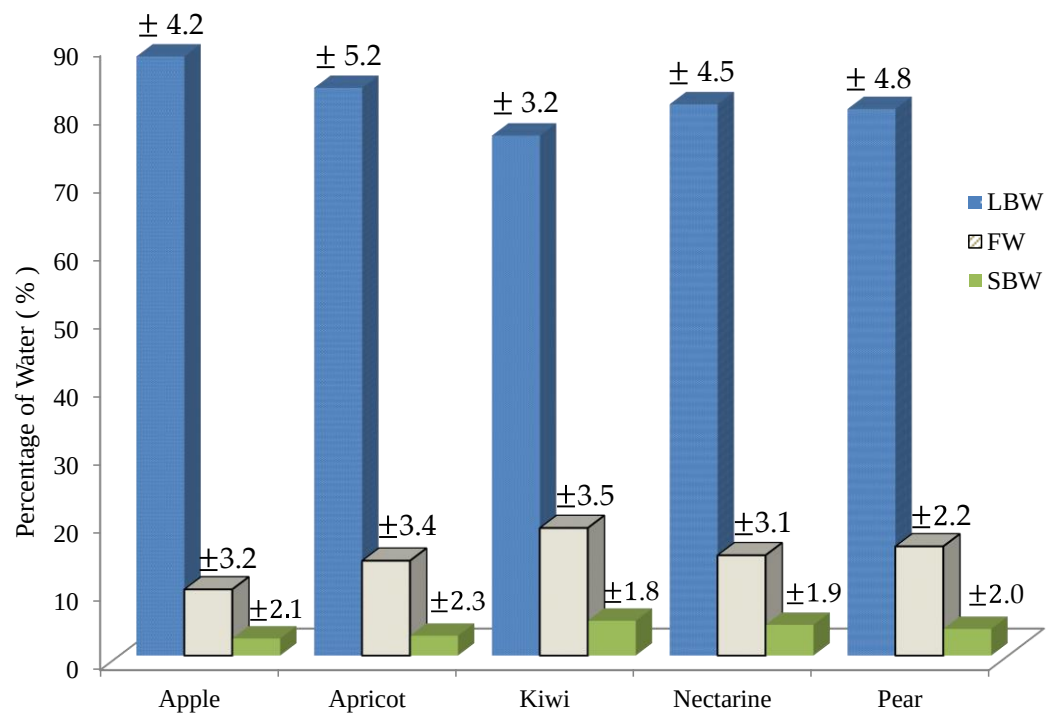


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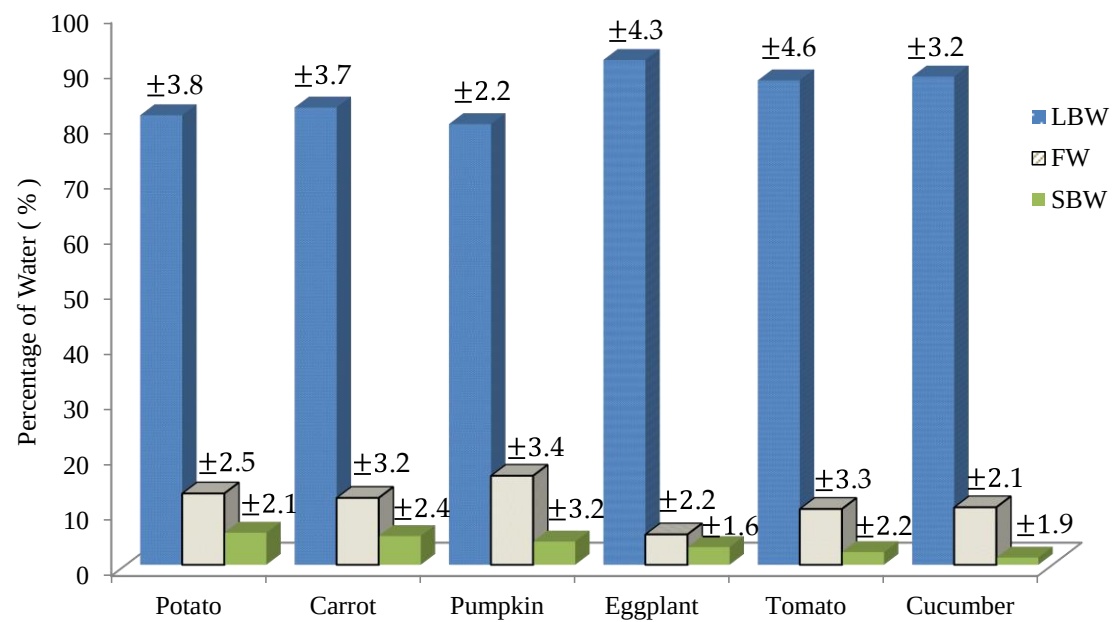


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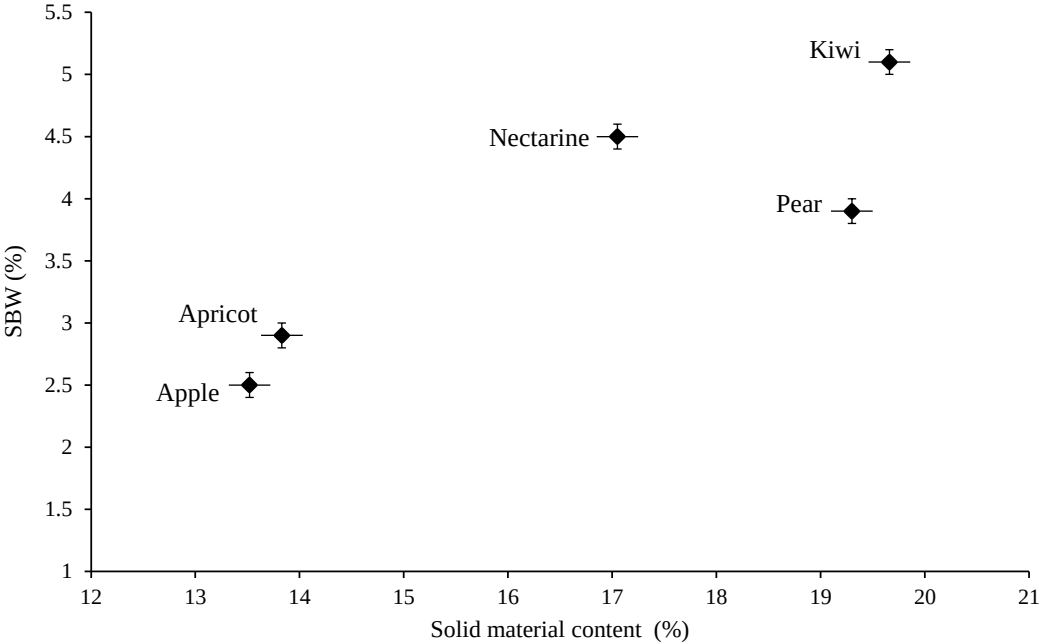


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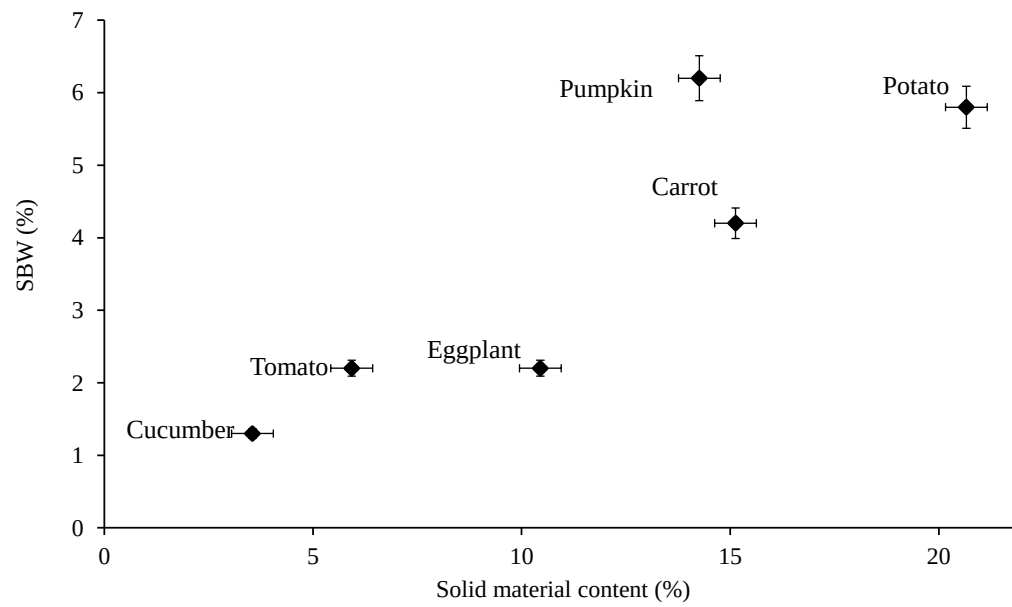


Figure 5(a):

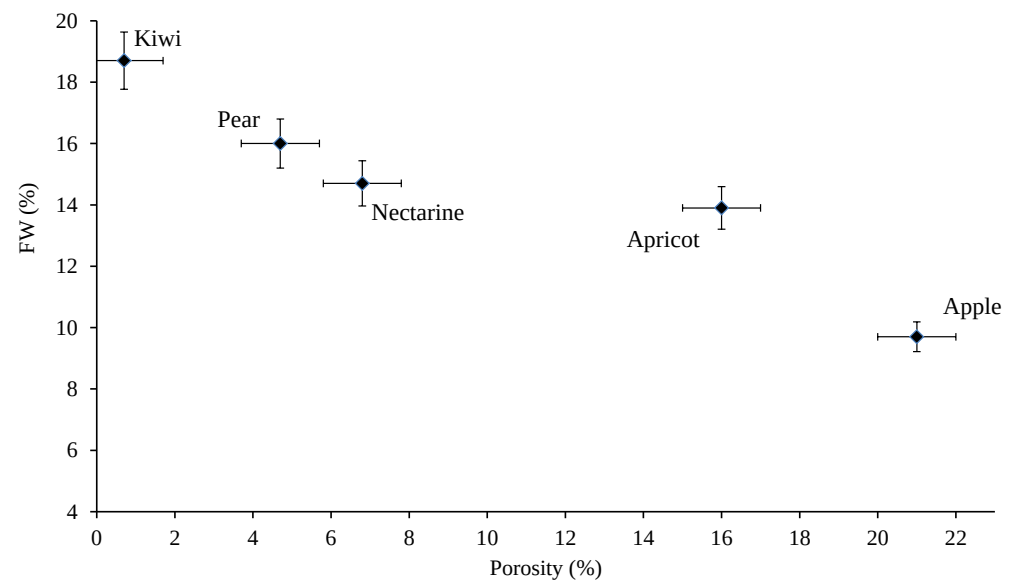


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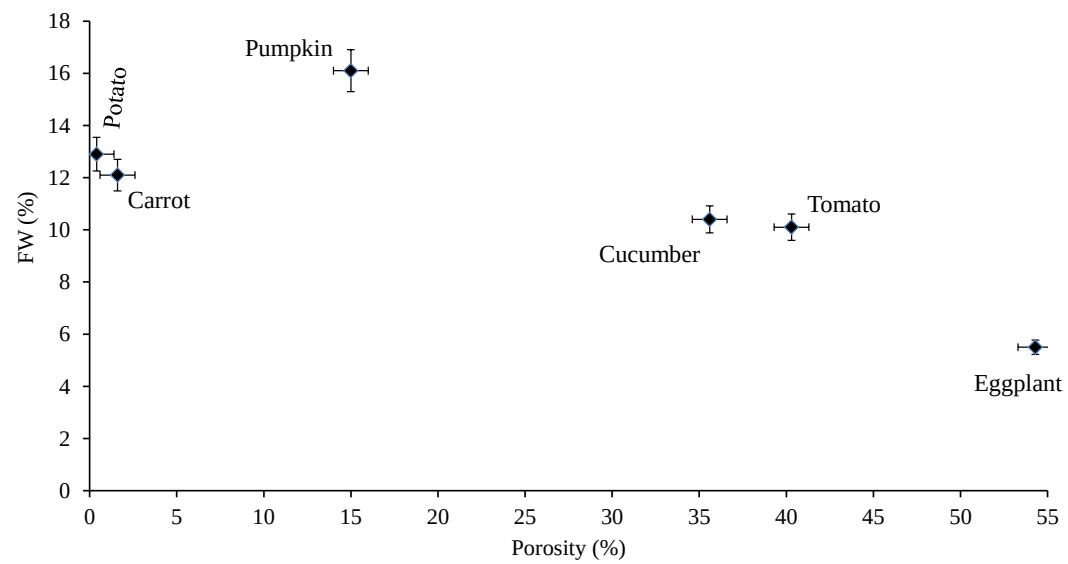


Figure 6:

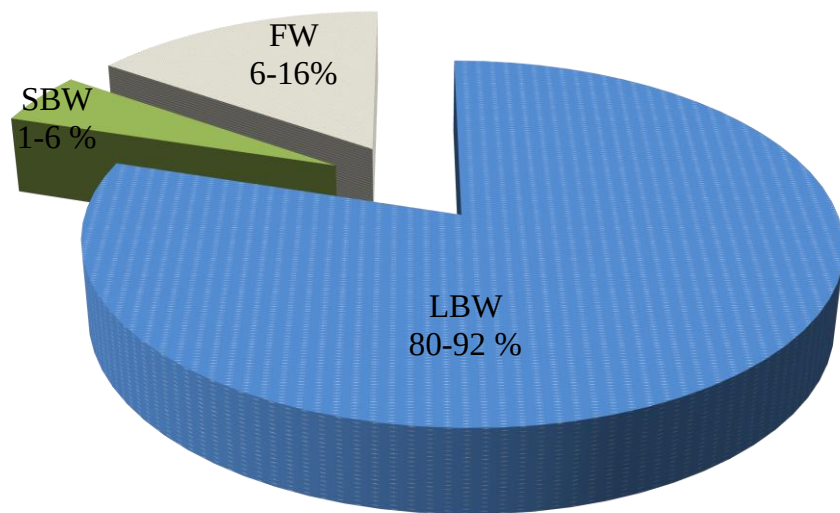


Table 1: The different T2 component of fruits and vegetables

Material	Slow component		Medium component		Fast component		Goodness of fit (R^2)
	T ₂ (ms)	%	T ₂ (ms)	%	T ₂ (ms)	%	
Apple	116.6 $\pm$ 12.2	87.8 $\pm$ 4.2	36.1 $\pm$ 6.6	9.7 $\pm$ 3.2	8.5 $\pm$ 3.6	2.5 $\pm$ 2.1	0.9993
Pear	156.3 $\pm$ 13.1	80.1 $\pm$ 4.8	35.4 $\pm$ 5.3	16 $\pm$ 2.2	4.7 $\pm$ 2.7	3.9 $\pm$ 2.0	0.9985
Kiwi	145.5 $\pm$ 14.2	76.2 $\pm$ 3.2	55.5 $\pm$ 6.7	18.7 $\pm$ 3.5	11.2 $\pm$ 3.1	5.1 $\pm$ 1.8	0.9974
Nectarine	113.1 $\pm$ 11.7	80.8 $\pm$ 4.5	49.8 $\pm$ 4.6	14.7 $\pm$ 3.1	9.6 $\pm$ 2.6	4.5 $\pm$ 1.9	0.9983
Apricot	117.8 $\pm$ 10.6	83.2 $\pm$ 5.2	46.7 $\pm$ 5.6	13.9 $\pm$ 3.4	7.2 $\pm$ 3.7	2.9 $\pm$ 2.3	0.9996
Potato	141.0 $\pm$ 11.2	81.3 $\pm$ 3.8	35.9 $\pm$ 9.6	12.9 $\pm$ 2.5	7.3 $\pm$ 2.6	5.8 $\pm$ 2.1	0.9984
Eggplant	102.4 $\pm$ 12.7	91.3 $\pm$ 4.3	30.5 $\pm$ 6.9	5.5 $\pm$ 2.2	6.3 $\pm$ 2.1	3.2 $\pm$ 1.6	0.9988
Pumpkin	137.0 $\pm$ 13.6	79.7 $\pm$ 2.2	65.4 $\pm$ 8.4	16.1 $\pm$ 3.4	12.6 $\pm$ 3.9	4.2 $\pm$ 3.2	0.9982
Tomato	118.6 $\pm$ 14.2	87.6 $\pm$ 4.6	32.2 $\pm$ 7.7	10.1 $\pm$ 3.3	5.5 $\pm$ 2.6	2.3 $\pm$ 2.2	0.9995
Cucumber	168.2 $\pm$ 15.2	88.3 $\pm$ 3.2	25.2 $\pm$ 5.6	10.4 $\pm$ 2.1	4.5 $\pm$ 2.8	1.3 $\pm$ 1.9	0.9964
Carrot	142.2 $\pm$ 11.2	82.7 $\pm$ 3.7	39.2 $\pm$ 6.9	12.1 $\pm$ 3.2	5.2 $\pm$ 2.6	5.2 $\pm$ 2.4	0.9976

